Lecture-18 [06 October 2026]

Temporal Differencing Methods for estimating V_∞^ϕ

Some basics: Let $\phi: \mathcal{S} \to \Delta(\mathcal{A})$ be given, and let $\pi = \{\phi, \phi, \cdots\}$.

* $V_\infty^\phi(s) = \mathbb{E}^\pi\left[\sum_{n=1}^{N} R_n \mid S_0 = s\right]$, $s \in \mathcal{S}$.

* The operator $L_\phi : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ defined as

$$L_\phi \underline{v} = r_\phi + P_\phi \underline{v}, \quad \underline{v} \in \mathbb{R}^{|\mathcal{S}|}$$

admits a unique fixed point equal to V_∞^ϕ.

Stochastic Approximation: The underlying principle for TD methods

Suppose X is a RV with mean μ. One way to estimate μ is to consider $X_1, \cdots, X_n \overset{\text{IID}}{\sim} F_X$, and set

$$\hat{\mu}_n = \frac{1}{n}\sum_{k=1}^{n} X_k. \qquad \text{(Sample mean)}.$$

By SLLN, $\hat{\mu}_n \xrightarrow[a.s.]{} \mu$ as $n \to \infty$. Robbins and Monro (1951) showed the following more general result.

$$\hat{\mu}_{n+1} = \frac{n}{n+1}\hat{\mu}_n + \frac{1}{n+1} X_{n+1}$$

Theorem (Robbins-Monro)

Let $\{\tau_n\}_{n \geq 1}$ be a non-negative sequence satisfying

$$\sum_{n=1}^{\infty} \tau_n = +\infty, \qquad \sum_{n=1}^{\infty} \tau_n^2 < +\infty,$$

and let $X_1, X_2, \cdots \overset{iid}{\sim} F_X$ with common mean $\mu \in \mathbb{R}$. Let the sequence $\{\hat{\mu}_n\}_{n \geq 1}$ be defined recursively via

$$\hat{\mu}_{n+1} = \hat{\mu}_n + \tau_n\left(X_n - \hat{\mu}_n\right), \qquad \hat{\mu}_1 \leftarrow \text{arbitrary value}.$$

Then,

$$\hat{\mu}_n \xrightarrow[a.s.]{} \mu \quad \text{as } n \to \infty.$$

- TD(0)
  - Offline
  - Online
- TD(γ)
  - Offline
  - Online

Offline TD(0)

To estimate $v_\infty^\phi(s)$ for a fixed $s \in \mathcal{S}$:

* Generate multiple episodes starting from $S_0 = s$ and implementing ϕ.
* For each $n \in \mathbb{N}$, let $h_n = \sum_{t=1}^{N_n} R_t$ denote the sum of rewards generated in the nth episode. Here, N_n = stopping time of nth episode.
* Set

$$V_{n+1}(s) = V_n(s) + \tau_n \left(h_n - V_n(s) \right) \quad , \quad V_0(s) \leftarrow \text{arbitrary initialisation.}$$

Remarks on Offline TD(0)

* The above offline variant requires waiting till one full episode is completed before an update can be made.
* It wastes considerable within-episode state information
* A separate estimate must be obtained for each state $s \in \mathcal{S}$.

Online TD(0)

It can be shown that v_∞^ϕ satisfies the one-step relation

$$\begin{aligned} v_\infty^\phi(s) &= \mathbb{E}^\pi \left[r(S, A, S') + v_\infty^\phi(S') \,\middle|\, S = s \right] \\ &= \mathbb{E}^\pi \left[r(s, A, S') + v_\infty^\phi(S') \right], \qquad \pi = \{\phi, \phi, \dots\}. \end{aligned}$$

A recursive way of estimating v_∞^ϕ using one step updates is

$$(s, a) \longrightarrow (s', r)$$

$$V_{n+1}(s) = V_n(s) + \tau_n \underbrace{\left(r(s, a, s') + V_n(s') - V_n(s) \right)}_{\text{temporal difference} =: \delta_n}$$

Remarks on Online TD(0)

* Usage of $V_n(s')$ in the highlighted term above is referred to as bootstrapping in RL literature
* The online TD(0) algorithm requires the knowledge of $V_n(s')$ for all $s' \in \mathcal{S}$ which are obtainable from $(s, a) \to s'$ transition

* In algorithmic form, online TD(0) is as follows.

Algorithm 10.3 : TD(0) for Estimating v_∞^ϕ in an Episodic Model

1. **Initialise :**
   a) Specify $\phi : \mathcal{S} \to \Delta(\mathcal{A})$.
   b) Specify number of replicates (episodes) K.
   c) Specify $\{\tau_n\}_{n \geq 1}$.
   d) $v(s) \leftarrow 0$ for all $s \in \mathcal{S}$.
   e) $\text{count}(s) \leftarrow 0$ for all $s \in \mathcal{S}$.
   f) $k \leftarrow 1$

2. **Iterate :** While $k \leq K$:
   a) Specify $s \in \mathcal{S} \setminus \Delta$.
   b) **Generate an episode :** While $s \notin \Delta$:
      i) $\text{count}(s) \leftarrow \text{count}(s) + 1$.
      ii) Sample $a \sim \phi(\cdot|s)$.
      iii) Simulate $(s', r) \longrightarrow$ (model-free, P, r unknown)
      OR
      (if P, r are known) $\longleftarrow$ Sample $s' \sim P(\cdot|s,a)$ and set $R \leftarrow r(s,a,s')$
      iv) Evaluate the temporal difference
      $$\delta \leftarrow R + v(s') - v(s)$$
      v) Set $v(s) \leftarrow v(s) + \tau_{\text{count}(s)}\, \delta$
      vi) $s \leftarrow s'$.
   c) $k \leftarrow k+1$.

3. **Terminate :** Return $v(s)$ for all $s \in \mathcal{S}$.

* The value estimate updates for different states are based on different learning rates. This is to account for the fact that each state may be visited different number of times. This is why $\tau_{\text{count}(s)}$ appears.
  → asynchronous updates

* Common choices of learning rates :

- Polynomial : $\tau_n = \alpha n^{-\beta}$, $\beta \in (\frac{1}{2}, 1]$, $\alpha > 0$.
- Ratio : $\tau_n = \frac{\alpha}{\beta + n}$, $\alpha > 0$, $\beta \geq 0$
- Logarithmic : $\tau_n = \alpha \frac{\log(n+1)}{n}$, $\alpha > 0$.

TD(γ) : Motivation

Observe that

$$v_\infty^\phi(s) = \mathbb{E}^\pi \left[r(S, A, S') + v_\infty^\phi(S') \mid S = s \right].$$

$$= \mathbb{E}^\pi \left[r(S, A, S') + r(S', A', S'') + v_\infty^\phi(S'') \mid S = s \right],$$

where $S \xrightarrow[A]{} S' \xrightarrow[A']{} S''$.

This suggests that we may use a 2-step update for $v(s)$ given by

$$v(s) \leftarrow v(s) + \tau \left(r(s, a, s') + r(s', a', s'') + v(s'') - v(s) \right)$$

$$= v(s) + \tau \left(\underbrace{r(s, a, s') + v(s') - v(s)}_{\delta} + \underbrace{r(s', a', s'') + v(s'') - v(s')}_{\delta'} \right)$$

$$= v(s) + \tau(\delta + \delta').$$

Alternatively, this may be viewed as two 1-step updates :

* Step 1 : $v(s) \leftarrow v(s) + \tau_1 \delta$ (keep $v(\cdot)$ unchanged for other states)

* Step 2 : $v(s) \leftarrow v(s) + \tau_2 \delta'$, $v(s') \leftarrow v(s') + \tau_2 \delta'$, keep $v(\cdot)$ unchanged for other states

In Online TD(γ), the second step update for $v(s)$ is modified as

$$v(s) \leftarrow v(s) + \tau_2 \gamma \delta', \qquad v(s') \leftarrow v(s') + \tau_2 \delta',$$

for some $\gamma \in [0, 1)$.

The above exercise can be repeated for m steps (starting from decision epoch n):

$$v_\infty^\phi(s) = \mathbb{E}^{\{\phi, \phi, \ldots\} \leftarrow \pi} \left[\underbrace{\sum_{k=0}^{m-1} r(S_{n+k}, A_{n+k}, S_{n+k+1}) + v_\infty^\phi(S_{n+m})}_{R_n^m} \mid S_n = s \right]$$

$v_\infty^\phi(s) = \sum_{m=1} p_m \alpha_m$ $\qquad$ $v_\infty^\phi(s) = \alpha_1$, $v_\infty^\phi(s) = \alpha_2$

That is, for any fixed decision epoch $n \in \mathbb{N}$ and $s \in \mathcal{S}$,

$$v_\infty^\phi(s) = \mathbb{E}^\pi\left[R_n^m \mid S_n = s\right] \qquad \forall\, m \in \mathbb{N}.$$

Then, we may write

$$v_\infty^\phi(s) = \sum_{m=1}^{\infty} \gamma^{m-1}(1-\gamma)\, \mathbb{E}^\pi\left[R_n^m \mid S_n = s\right]$$

$$= (1-\gamma) \sum_{m=1}^{\infty} \gamma^{m-1} \sum_{k=0}^{m-1} \mathbb{E}\left[r(S_{n+k}, A_{n+k}, S_{n+k+1}) + v_\infty^\phi(S_{n+m}) \mid S_n = s\right]$$

$$= (1-\gamma) \sum_{k=0}^{\infty} \sum_{m=k+1}^{\infty} \gamma^m\, \mathbb{E}\left[r(S_{n+k}, A_{n+k}, S_{n+k+1}) + v_\infty^\phi(S_{n+m}) \mid S_n = s\right].$$

It can be shown, after some algebra, that

$$v_\infty^\phi(s) = v_\infty^\phi(s) + \sum_{m=0}^{\infty} \gamma^m\, \mathbb{E}^\pi\left[r(S_{n+m}, A_{n+m}, S_{n+m+1}) + v_\infty^\phi(S_{n+m+1}) - v_\infty^\phi(S_{n+m}) \mid S_n = s\right]$$

Offline TD(γ)

Starting from an initial state, generate a full trajectory

$$S_0, a_0, S_1, a_1, \ldots, S_n, a_n, S_{n+1}, \ldots$$

until the completion of episode. Update

$$v(S_n) \leftarrow v(S_n) + \tau \sum_{m=0}^{\infty} \gamma^m \Big(r(S_{n+m}, a_{n+m}, S_{n+m+1}) + v(S_{n+m+1}) - v(S_{n+m}) \Big)$$

$$= v(S_n) + \tau \sum_{m=0}^{\infty} \gamma^m\, \delta_{n+m},$$

where

$$\delta_k = r(S_k, a_k, S_{k+1}) + v(S_{k+1}) - v(S_k).$$

Remarks :

* Once a full trajectory (episode) is completed, the above recursive updates are done for each of the states visited in the trajectory.
* τ is just a placeholder for τ_n or $\tau_{\text{Count}(s)}$
* TD(0) is same as TD(γ) with $\gamma = 0$.

* The upper limit of the summation $\sum_{m=0}^{\infty} \gamma^m \delta_{n+m}$ is usually restricted to the length of the trajectory. So, the sum can be evaluated.

* If $\gamma=1$, then the update becomes

$$V(s_n) \leftarrow V(s_n) + \tau \sum_{m=0}^{\infty} \delta_{m+n}$$

$$= V(s_n) + \tau \sum_{m=n}^{\infty} \delta_m$$

$$= V(s_n) + \tau \sum_{m=n}^{\infty} \left(r(s_m, a_m, s_{m+1}) + V(s_{m+1}) - V(s_m) \right)$$

$$= (1-\tau) V(s_n) + \tau \sum_{m=n}^{\infty} r(s_m, a_m, s_{m+1}).$$

Observe that

$$\sum_{m=n}^{\infty} r(s_m, a_m, s_{m+1})$$

is the Monte-Carlo estimate of the total reward starting from $S_n = s_n$.

Thus, TD(1) = a variant of Monte-Carlo!

Online TD(γ) and Eligibility Traces

In an online version of TD(γ), once s_n is observed at the nth epoch, the value estimates of all previously observed states are also updated in addition to the update for $V(s_n)$ as follows:

$$V(s_n) \leftarrow V(s_n) + \tau_n \delta_n$$

$$V(s_{n-1}) \leftarrow V(s_{n-1}) + \tau_n \gamma \delta_n$$

$$V(s_{n-2}) \leftarrow V(s_{n-2}) + \tau_n \gamma^2 \delta_n$$

$$\vdots$$

$$V(s_1) \leftarrow V(s_1) + \tau_n \gamma^{n-1} \delta_n$$

$$V(s_0) \leftarrow V(s_0) + \tau_n \gamma^n \delta_n.$$

If s_n is visited multiple times in the trajectory, its value estimate gets updated multiple times. This can be avoided by updating the value estimate for s_n only once, even if it has occurred multiple times.

Eligibility traces

An eligibility trace is a collection of functions on $\mathcal{S}$ of the following form:

$$\{e_n\}_{n\geq 1}, \quad e_n : \mathcal{S} \to \mathbb{R}$$

Two variants are introduced in the literature:

a) Accumulation trace:

$$e_1(s) = \mathbb{1}_{\{s_1\}}(s) = \begin{cases} 1, & s = s_1, \\ 0, & s \neq s_1. \end{cases}$$

$$\forall n \geq 2: \quad e_n(s) = \begin{cases} \gamma e_{n-1}(s) + 1, & s = s_n, \\ \gamma e_{n-1}(s), & s \neq s_n. \end{cases}$$

Using the accumulation trace, Online TD(γ) updates may be written succinctly as

$$V(s) \leftarrow V(s) + \tau_n \delta_n e_n(s)$$

(Eg: $V(s_n) \leftarrow V(s_n) + \tau_n \delta_n e_n(s_n)$

$= V(s_n) + \tau_n \delta_n + \tau_s \delta_n \gamma e_{n-1}(s_n)$)

On the other hand, replacement trace is defined by

$$e_1(s) = \mathbb{1}_{\{s_1\}}(s), \qquad e_n(s) = \begin{cases} 1, & \text{if } s = s_n, \\ \gamma e_{n-1}(s), & \text{if } s \neq s_n. \end{cases}$$

Algorithm 10.4: TD(γ) for estimating v_∞^ϕ for an Episodic Model

1. Initialise:

a) Specify $\phi : \mathcal{S} \to \Delta(\mathcal{A})$.
b) $V(s) \leftarrow 0 \ \forall s$, $e(s) \leftarrow 0 \ \forall s$, $\text{Count}(s) \leftarrow 0 \ \forall s$
c) Specify $\gamma \in [0,1]$, $\{\tau_n\}_{n\geq 1}$
d) Specify number of episodes K.
e) Set $k \leftarrow 1$.

2. Iterate: While $k \leq K$:

a) Specify $s \in \mathcal{S} \setminus \Delta$.

b) Generate an episode: While $s \notin \Delta$:

i) $e(s) \leftarrow e(s) + 1$ (accumulation trace)
$e(s) \leftarrow 1$ (replacement trace)

ii) $\text{count}(s) \leftarrow \text{count}(s) + 1$

iii) Sample $a \sim \phi(\cdot|s)$

iv) Simulate (s', r)
OR
Sample $s' \sim P(\cdot|s, a)$ and set $R \leftarrow r(s, a, s')$.

v) Evaluate the temporal difference

$$\delta \leftarrow R + v(s') - v(s).$$

vi) For all $s \in \mathcal{S}$,

$$v(s) \leftarrow v(s) + \tau_{\text{count}(s)}\, \delta\, e(s)$$

and

$$e(s) \leftarrow \gamma\, e(s)$$

vii) $s \leftarrow s'$.

c) $k \leftarrow k+1$

3. Terminate: Return $v(s)$ for all $s \in \mathcal{S}$.